

Global Facilities - Design & Construction Standard - CSA - FAB Building and Cleanroom Interior

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[ECN History](#)[Submit a Change Request](#)

1.0 Purpose

- 1.1 This document provides a Required Standard and Standard Guideline on material and design specification that shall be followed and complied in the Design, Engineering and Construction of FAB Building and Cleanroom Interior as part of any green and/or brown field construction projects contract requirement.
- 1.2 The standard is applicable for any green and/or brown field construction projects with regards to provide FAB Building and Cleanroom Interior as per required in the contract.
- 1.3 The document is intended to serve as a reference and guideline in addition to any Construction Drawings and Specification provided in the contract requirement including any required by local authority regulation and standard requirement as well as code of practice.
- 1.4 The established Specifications Standard for each locale shall also comply with Micron's in-house:
 - 1.4.1 RBA Code of Conduct requirements for Semiconductor Manufacturing
 - 1.4.2 Environmental, Health & Safety Standards
 - 1.4.3 Operational Requirements /SOP (Standard Operation Procedures)
 - 1.4.4 Security Protocol
- 1.5 The Specifications Standard to be adopted in each local shall not violate any local codes or by-laws and reflects the current semi-conductor industry's best practices.

- 1.6 Where there is the option to adopt either the MasterSpec or in-region specifications standard, the AE (Architectural & Engineering Consultants) shall propose the most suitable solution based on safety, quality, time & cost to Micron's Designated Project Team (DPT) for review and acceptance. Otherwise, the more stringent option shall apply.
- 1.7 The loadings listed in this document are standard loadings. Loadings may vary as per local code requirements versus Micron standard loadings, and each project shall use whichever is higher (stricter) value for design.

2.0 Scope

- 2.1 This document for FAB Building and Cleanroom Interior CSA specification covers all necessary related design, engineering, procurement, construction, installation and completion of piling, substructure, superstructure, and roof structure to full compliance of the contract requirement and required for the project. It extended to cover any green and/or brown field construction projects.
- 2.2 This document applies to all Micron Facilities employees, contractors, and vendors at all Micron sites.

3.0 Definitions and Acronyms

3.1 Definitions of acronyms used in this document are listed as follows:

- **AE** – External Architectural & Engineering Consultants
- **AMHS** – Automated Material Handling System
- **DCC** – Dry Cooling Coil
- **FAT** – Factory Acceptance Test
- **FCT** – Facilities Central Team. This is used interchangeably with Global Facilities Engineering Team and associated to O&M BKM
- **FFU** – Fan Filter Unit
- **GFTT** – Global Facilities Technology Team. This is used interchangeably with Global Facilities Engineering Team and associated to design standards.
- **GFC** – Global Facilities & Construction Team. This is used interchangeably with Global Facilities (GFac), Facilities Central Group (FCG), World Wide Operations (WWOps), World Wide Facilities (WW_FACILITIES), Corp Facilities
- **Global Facilities Central Team** – This is used interchangeably with Innovation & Integration Team
- **Global Facilities Construction Team** – This is used interchangeably with Global Construction Team
- **Global Facilities Operations Support Team** – This is used interchangeably with Global Facilities Operations Team
- **Global Facilities Planning Team** – This is used interchangeably with Global Facilities Line Planning Team
- **HEPA** – High Efficiency Particulate Arrest (Filter)
- **IST** – Integrated System Test
- **ITP** – Inspection and Test Plan
- **LOTO/COHE** - Lockout Tagout / Control of Hazardous Energy
- **MAU** - Makeup Air Unit
- **OAT** – Operational Acceptance Test
- **QP/PE** - External Qualified Person (professional architect/engineer)
- **RA** – Return Air
- **RAC** – Return Air Chase
- **SA** – Supply Air
- **SAT** – Site Acceptance Test
- **ULPA** – Ultra Low Penetration Air (Filter)
- **VC** – Vibration Criterion (Curve)

4.0 Roles and Responsibilities

Roles	Responsibilities
GFTT Lead (Discipline Engineer)	• Develop and maintain System Standard document • Ensure Best Known Methods (BKM), lessons learned, and emerging technologies are documented in System Standard • Perform periodic document review and update System Standard
FCT Lead (Discipline Engineer)	• Review and provide technical feedback on System Standard • Support the Site teams by providing knowledge and technical expertise
Global Construction and Site Project Construction Teams	• Review and provide technical feedback on System Standard • Incorporate System Standard into Project Engineering, Design, and Construction • Document project deviations, exceptions, and exclusions from System Standard • Share new lessons learned to GFTT Lead, to determine if it is Best Known Method (BKM) that can be captured in future revision of System Standard
Site Facilities Teams	• Use System Standard as technical reference

5.0 Cleanroom Vibration Control Design

Vibration control is critical for quality particle control, lab tool resolution, and high precision and metrology tools. This specification highlights Micron's vibration requirement or criteria for Facility, AT equipment, and AMHS. Micron develop, consolidate, and formalize global specifications in terms of building vibration requirements, mitigation guidelines, and validation process.

Understanding environmental and design loading requirements is an important element when designing and building a FAB cleanroom. Vibration control design involves working with multi-disciplinary design team of architect, civil & structural engineers, mechanical, electrical, process engineers, and cleanroom specialists.

Micron Cleanroom Front and Back End facilities have different vibration requirements. Precision reading and monitoring are essential to keep the equipment and condition up to the required standards for cleanroom operations. Vibration is the most significant factors that can compromise the design of the entire FAB building. Micron to engage a vibration consultant to effectively address and resolve the challenges in the design of the FAB Cleanroom.

1. For concept design of a new location, the vibration consultant should conduct a field investigation by installing a monitoring system to observe the ground vibrations within the tract and analyze/understand the response characteristics of all external activities such as roads, nearby freight and passenger trains, or large mechanical vehicular activities, earthquakes, blast sites, airport runways, future MRT projects, military buildings, etc. The vibration consultant to submit the final report which details the study, findings, and recommendations to Micron to be used as reference by the structural consultant when designing the main structure of the FAB building.
2. As the vibration response of a structure are influenced by its dynamic stiffness and damping, for structural layout design of the new or extension of FAB Cleanroom level should be designed with high dynamic stiffness aside from the total loading requirements. **Front End Cleanroom** is recommended to use precast or cast-in situ waffle slab and for **Back End Cleanroom** to use precast double or single tee slab. Hollow core slab for **Back End Cleanroom** is not advisable as it is not stiffened enough against the vibration requirements. Support Equipment's for cleanroom should be in a separate building connected the production or cleanroom area by using expansion joint. Subfab or lower levels should be supported by rigid foundation and both cleanroom and support buildings should not be combined and demarcated by the expansion joint.

General Vibration Design Guides:

- American Institute of Steel Construction (AISC) Steel Design Guide Series 11
- National Institute of Health (NIH) General Design Guidelines, Section 4.E.3.7.1
- American Society of Heating, Refrigerating and Air-conditioning Engineers (ASHRAE) Handbook, Chapter 47
- Department of Defense Unified Facilities Criteria (UFC), Section 2-3-c

6.0 Cleanroom Vibration Loads

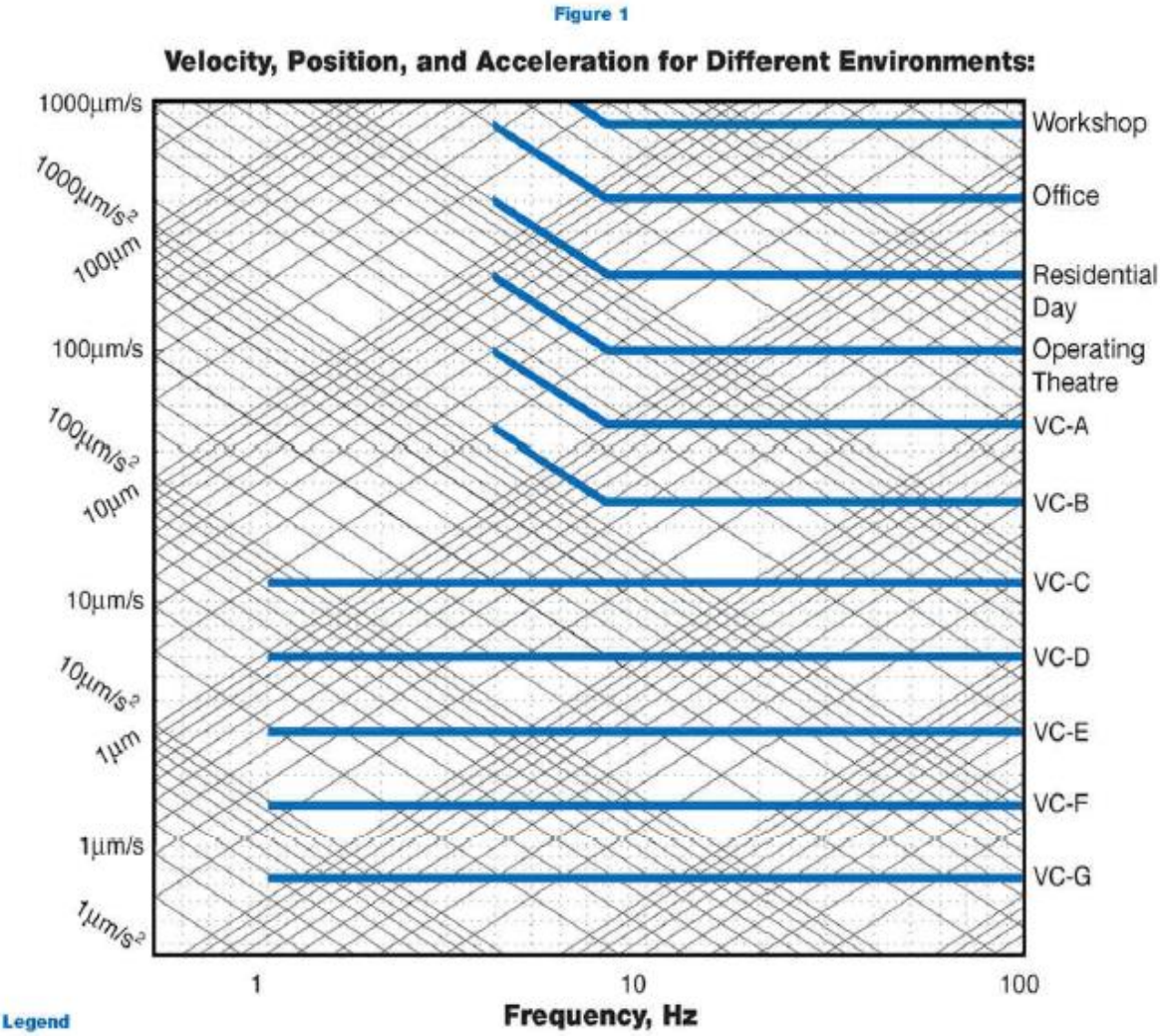
Micron VC floor vibration guidelines for cleanroom areas are based on root-mean-square vibration velocity in the $1/3^{\text{rd}}$ octave frequency spectra. The generic vibration criteria curves are the industry standard used to aid structural engineers in the design of Fab cleanroom buildings. Organizations such as AISC, ASHARE, IEST, and NIH recommend their use. These guidelines are valuable in the preliminary or concept design. Classifications range from "A" to "G" depending on the sensitivity. Where class A is the least stringent and Class G is the most stringent – see Table 6.1.

Micron specific vibration requirements for Front End Cleanroom, Photo and EUV Areas, Back End Cleanroom, and Probe is listed in **Table 7.1**.

6.1 Vibration Tabulated Cleanroom Requirements:

Criterion Curve	Amplitude ¹ μm/s (μin/s)	Detail Size ² μm	Description of use
Workshop (ISO)	800 (32,000)	N/A	Distinctly perceptible vibration. Appropriate to workshops and nonsensitive areas.
Office (ISO)	400 (16,000)	N/A	Perceptible vibration. Appropriate to offices and nonsensitive areas.
Residential day (ISO)	200 (8,000)	75	Barely perceptible vibration. Appropriate to sleep areas in most instances. Usually adequate for computer equipment, hospital recovery rooms, semiconductor probe test equipment, and microscopes less than 40X.
Operating theatre (ISO)	100 (4,000)	25	Vibration not perceptible. Suitable in most instances for surgical suites, microscopes to 100X and for other equipment of low sensitivity.
VC-A	50 (2,000)	8	Adequate in most instances for optical microscopes to 400X, microbalances, optical balances, proximity and projection aligners, etc.
VC-B	25 (1,000)	3	Appropriate for inspection and lithography equipment (including steppers) to 3μm line widths.
VC-C	12.5 (500)	1 - 3	Appropriate standard for optical microscopes to 1000X, lithography and inspection equipment (including moderately sensitive electron microscopes) to 1 μm detail size, TFT-LCD stepper/scanner processes.
VC-D	6.25 (250)	0.1 - 0.3	Suitable in most instances for demanding equipment, including many electron microscopes (SEMs and TEMs) and E-Beam systems.
VC-E	3.12 (125)	< 0.1	A challenging criterion to achieve. Assumed to be adequate for the most demanding of sensitive systems including long path, laser-based, small target systems, E-Beam lithography systems working at nanometer scales, and other systems requiring extraordinary dynamic stability.
VC-F	1.56 (62.5)	N/A	Appropriate for extremely quiet research spaces; generally difficult to achieve in most instances, especially cleanrooms. Not recommended for use as a design criterion, only for evaluation.
VC-G	0.78 (31.3)	N/A	Appropriate for extremely quiet research spaces; generally difficult to achieve in most instances, especially cleanrooms. Not recommended for use as a design criterion, only for evaluation.
¹ As measured in one-third octave bands of frequency over the frequency range 8 to 80 Hz (VC-A and VC-B) or 1 to 80 Hz (VC-C through VC-G) ² The detail size refers to line width in the case of microelectronics fabrication, the particle (cell) size in the case of medical and pharmaceutical research, etc. It is not relevant to imaging associated with probe technologies, AFMs, and nanotechnology. <i>The information given in this table is for guidance only. In most instances, it is recommended that the advice of someone knowledgeable about applications and vibration requirements of the equipment and processes be sought.</i>			

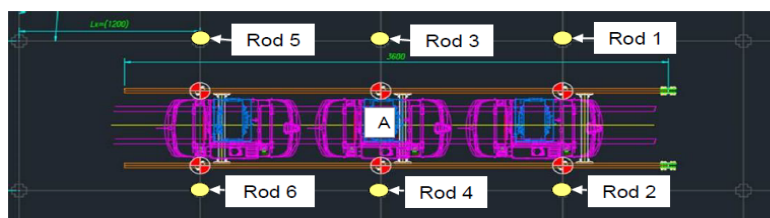
Reprinted with permission from Colin Gordon Associates. VCA–VCG refer to accepted standards for vibration sensitive tools and instruments. The levels displayed are RMS values measured in $1/3$ octave band center frequencies. $1/3$ octave plots are discussed in section 1.2.3.



6.2 Cleanroom Typical AMHS Loadings:

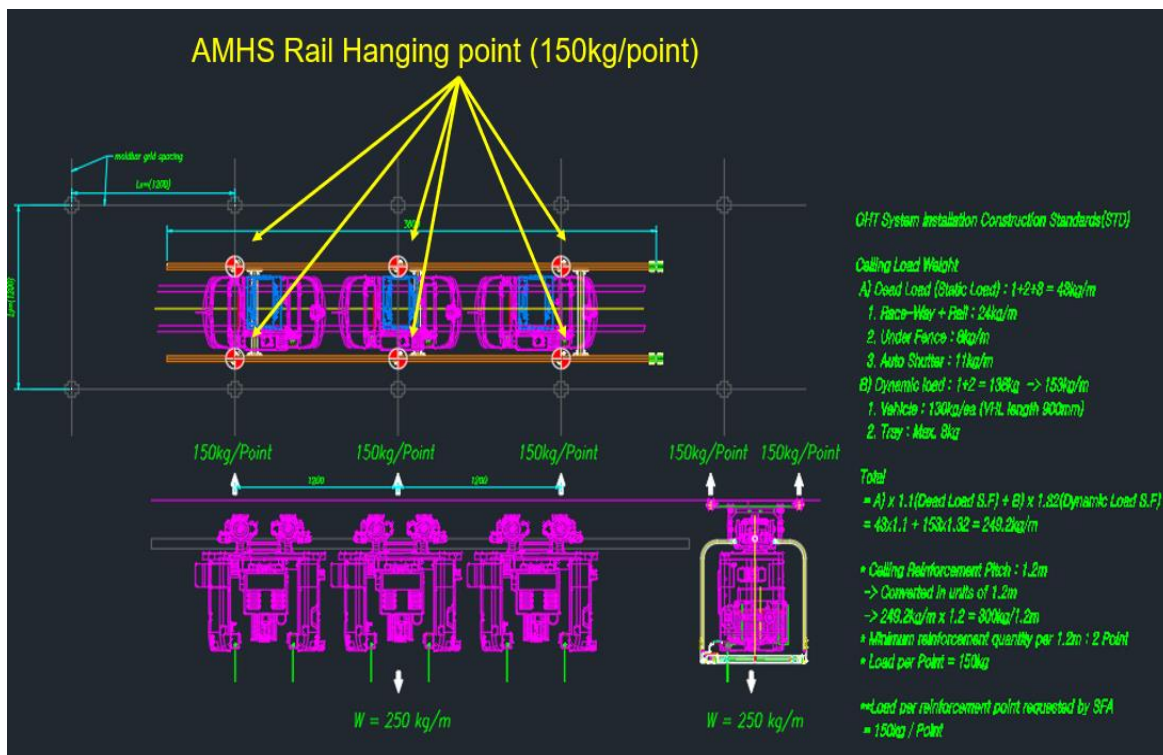
6.2.1 Back End (Assembly & Test) AMHS Loading & Vehicle Speed

Unless otherwise stated, the AMHS loading shall follow the minimum requirements below for average speed of vehicles range as 2.5m/s. Total Mass (Vehicle + Payload) to round off to 150 kg/point load to the bottom chord of the truss. Hence, the maximum speed can run up to maximum speed of 4.16m/s.



In this calculation only one (1) vehicle, i.e. vehicle A is considered passing Rod 3 and Rod 4. The other two vehicles in the figure above are excluded in the calculation.

- Total Mass = 130kg (Vehicle) + 8kg (Payload) = **138kg**
- Dynamic load on Rods 3 & 4 = $138 \times 9.8 = \mathbf{1353.8N}$
- Speed of vehicle = **2.5m/s**
- Time takes to travel from 1 grid line = $1.2/4.17 = \mathbf{0.48sec}$



Refer to the tabulated cleanrooms structural loadings below. Loadings may vary as per local code requirements versus Micron standard loading specifications and to use whichever is higher value for the design.

6.2.2 Front End AMHS Ceiling Loading & Height Requirements

Green Field Fab - General Standard (2022-)

Clean Room	Ceiling Height	Ceiling Loading (1.2m Ceiling Grid)	OHT System	OHT Type	Update	Building	Floor Level
Main Manufacturing (Ball Room)	6.7m	500 kg/sqm (max700kg/1 hanging point)	Dual OHT	G	07 Oct,2022	ID1, IDx, New Fab	Main C/R
EUV Room (FOUP Delivery)	6.7m	500 kg/sqm (max700kg/1 hanging point)	Dual OHT + ART	G	07 Oct,2022	ID1, IDx, New Fab	Main C/R
EUV Room (Reticle Delivery)	8.5m at ART space	300 kg/sqm (max500kg/1 hanging point)	Dual OHT + ART	G	07 Oct,2022	ID1, IDx, New Fab	Main C/R
Bonding Area	6.7m	500 kg/sqm (max700kg/1 hanging point)	Dual OHT	G	07 Oct,2022	ID1, IDx, New Fab	Main C/R
AMHS Maintenance Shop	6.7m	500 kg/sqm (max700kg/1 hanging point)	Dual OHT	G	07 Oct,2022	ID1, IDx, New Fab	Main C/R
AMHS Link Bridge	6.7m	500kg/sqm (max700kg/1 hanging point)	Dual OHT w/o STB	G	07 Oct,2022	New Fab	Main C/R
Probe	5.0m/6.7m	350kg/sqm (max500kg/1 hanging point)	Single OHT	G	07 Oct,2022	New Fab	Main C/R

Site Specific (including Retro Fit, and Yellow/Brown Field)

Clean Room	Ceiling Height	Ceiling Loading (1.2m Ceiling Grid)	OHT System	Update	Building	Floor Level
B51 Utility - Lifter Hub Upper AMHS Room	6.7m	500 kg/sqm (max700kg/1 hanging point)	Dual OHT (HW only)	G	07 Oct,2022	B51 Utility
B51 Utility - Lifter Hub Lower AMHS Room	4.8m	350kg/sqm (max500kg/1 hanging point)	Single OHT	SRC	07 Oct,2022	B51 Utility

6.3 Table 1. Cleanroom Level Super Imposed Dead Loads

Room Occupancy or Use	Uniform (kN/m2 or kPa)
1. All Clean Room Area	3.5
2. Clean Room EUV Area	3.5

6.4 Table 2. Cleanroom Level 03 Live loads

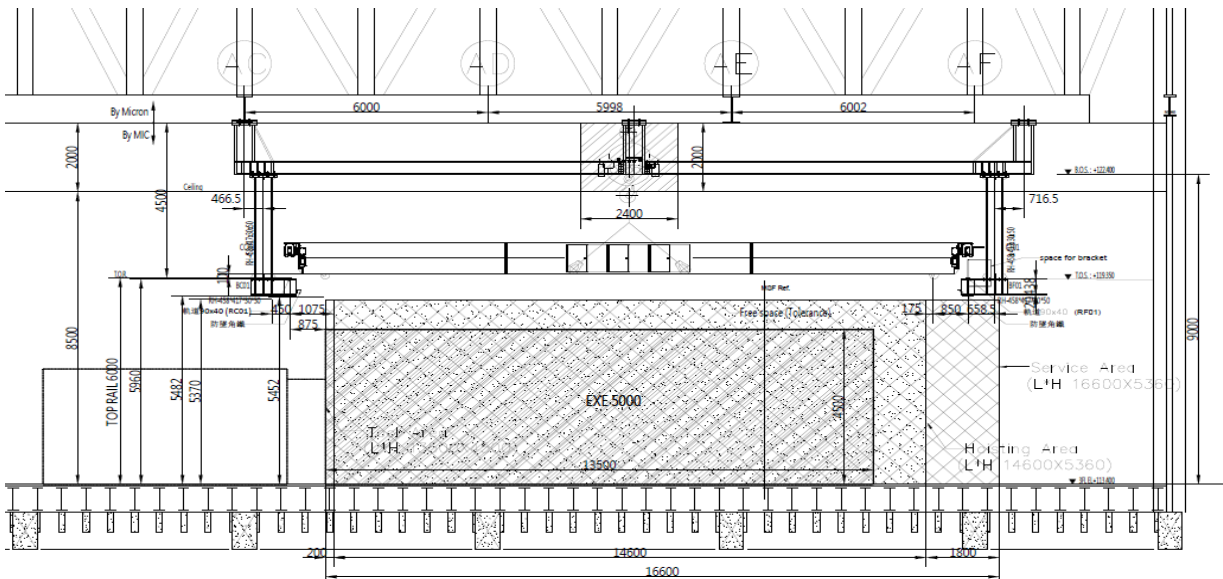
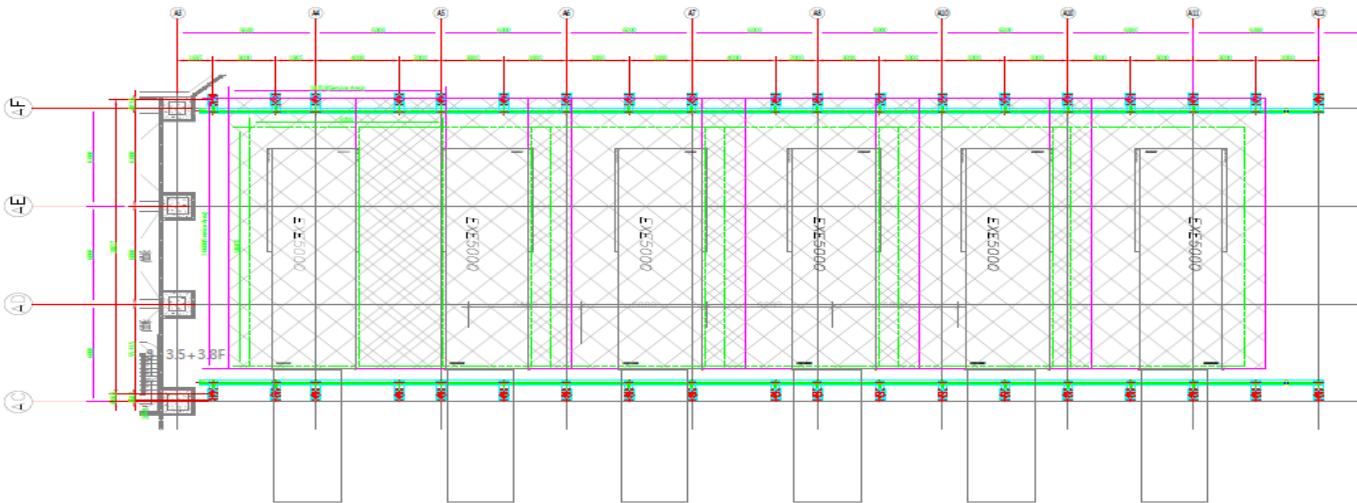
Room Occupancy or Use	Uniform (kN/m2 or kPa)
1. All Clean Room Area	25
2. Clean Room EUV Area	35

*For EUV Unit, the maximum mass per factory level is 147,000 kg excluding the support frame crane, piping and electrical cables and their supporting material.

6.5 EUV HiNA Crane Loads, 3D, Plan Layout, and Section Reference

The point load of each Crane is 47.35 tons, with the possibility of 2 cranes aligned side-by-side for lifting.

	EXE5XXX (HiNA)
Crane capability to support EUV scanner Type	All NXE/ EXE
Crane spam	17250mm
Point Load	47.35 tons



7.0 Cleanroom Vibration & Opening Requirements

7.1 Refer to the tabulated cleanroom minimum requirements.

Front End Cleanroom Areas (that are Non- Photo/Metrology/RDA and Non-EUV) Vibration Requirements	Criterion Curve	Amplitude (micro-m/s)	Amplitude (micro-in/s)
1. All Cleanroom Areas except Photo/Metrology/RDA and EUV – FAB Level	VC-C	12.5	500
Front End Photo/Metrology/RDA and EUV Vibration Requirements	Criterion Curve	Amplitude (micro-m/s)	Amplitude (micro-in/s)
1. All Photo/EUV/Met/RDA Area – FAB Level	VC-D+	5.08	200
2. Clean Subfab Level Photo/EUV Area	VC-C	12.5	500
Back End	Criterion Curve	Amplitude (micro-m/s)	Amplitude (micro-in/s)
1. CR1 DP Room Areas only	VC-D+	5.08	200
2. All Cleanroom Areas except CR1 DP	VC-C	12.5	500

For more detailed vibration requirements covering assembly equipment's process / packages, please refer to the Global Back End vibration [Link](#).

Probe	Criterion Curve	Amplitude (micro-m/s)	Amplitude (micro-in/s)
1. All Probe Areas	VC-C	12.5	500

7.2 Dynamic Stiffness Requirements for EUV Tools Requirements

Stiffness limits

A building suitable for the installation of the TWINSKAN EXE:5x00 platform must comply with the stiffness limit requirements below.

Static vertical stiffness at the Exposure Unit support points 4, 6, 7, and 8 of the Machine Support Frame must be > 5E+8 N/m.

Table 2.4 Stiffness limits at the interface level of the Exposure Unit to the Machine Support Frame^[a]

Frequency [Hz]	Dynamic stiffness Kxyz; measured at interface level of Exposure Unit (EU) MSF	Dynamic stiffness Kxyz; measured at interface level of the Vacuum Wafer Handler (VWH)	Dynamic stiffness Kxyz; measured at interface level of the Atmospheric Wafer Handler (AWH)	Dynamic stiffness Kxyz; measured at interface points for AWH	Dynamic stiffness Kxyz; measured at interface level of Front Support Module (FSM)	Dynamic stiffness Kxz; measured at interface level for FSM	Dynamic stiffness Ky; measured at interface level for FSM
	Point 4 to 10; 31 to 37	Point 15	Center-point 12	Point 13; 14	Point 101; 102; 104; 105	Point 2; 3	
	[N/m]	[N/m]	[N/m]	[N/m]	[N/m]	[N/m]	[N/m]
< 7	> 3E+8	> 3E+8	> 3E+8	> 3E+8	> 3E+8	> 3E+8	
[7-18]	> 5E+8	> 4E+8	> 4E+8	> 5E+8	> 5E+8	> 5E+8	> 4E+8
[18-50]		> 2.5E+8	> 2.5E+8	> 2.5E+8	> 2.5E+8	> 2.5E+8	
> 50	No requirement	No requirement	No requirement	No requirement	No requirement	No requirement	

^[a] If a Support Point position shown in Figure 2.5 is not indicated in the table, no requirements for stiffness or vibrations are applicable.

7.3 Minimum Cleanroom Openings

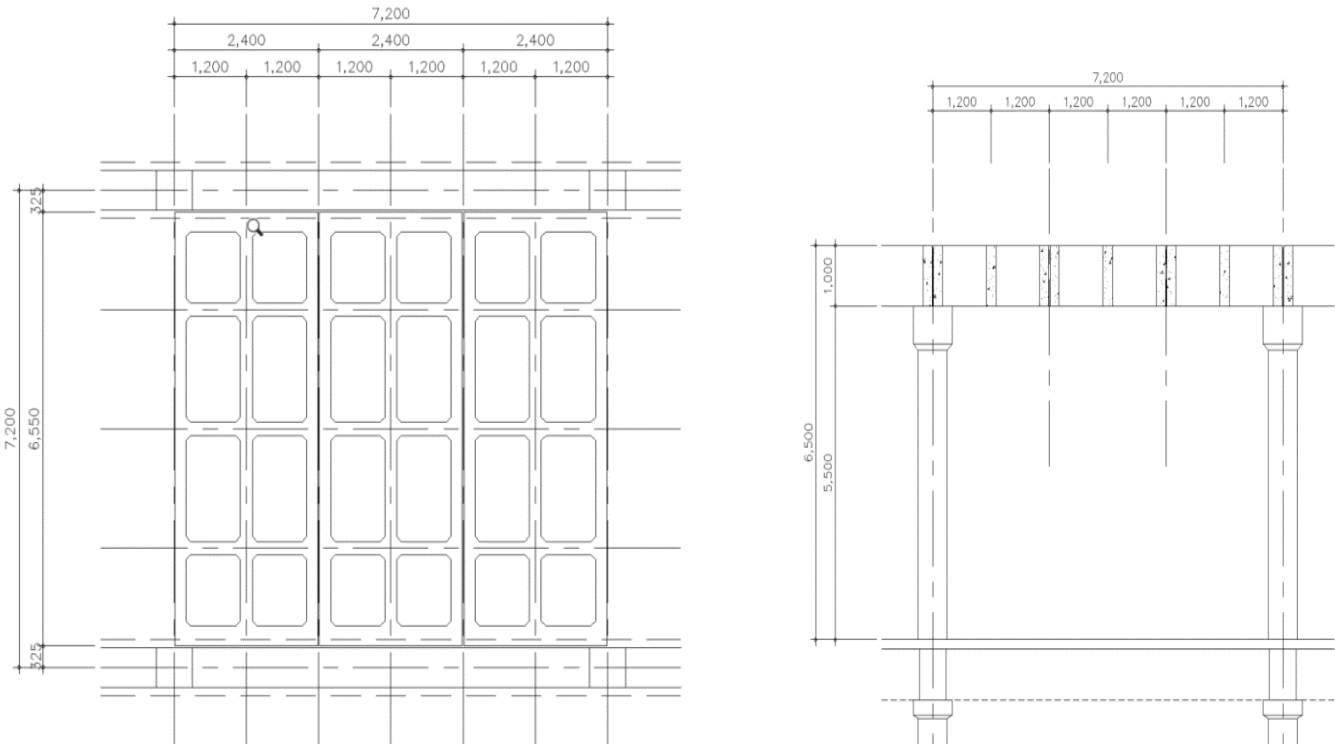
The minimum feed-through opening requirements for Front and Back end is less than or equal to 310mm.

Feed-Through Opening Requirements	Minimum (mm)
1. Waffle Slab (VC-D and VC-D+)	≥ 450
2. Double/Single Tee (VC-C)	≥ 310

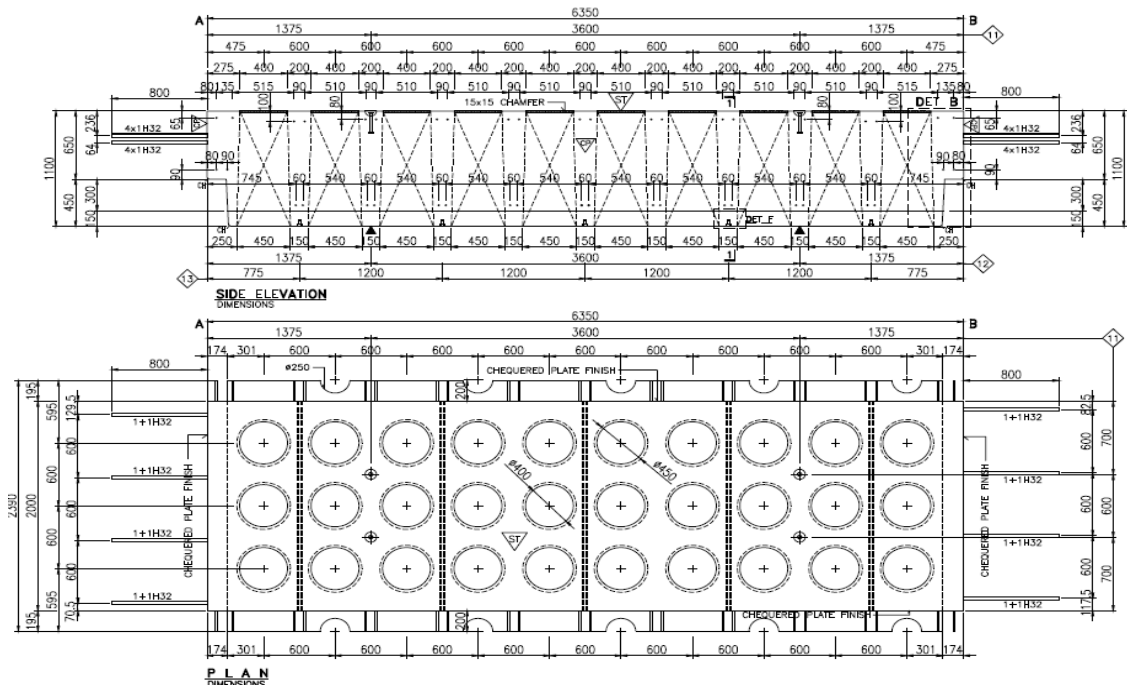
8.0 Cleanroom Structural Framing System

It depends on the region and availability of precast, waffle slab may either be precast or cast-in-situ. Below the typical precast element details. The depth varied from 900 to 1100mm depending on the structural design loading and vibration requirements.

8.1 Waffle Slab Rectangular – 3 Module

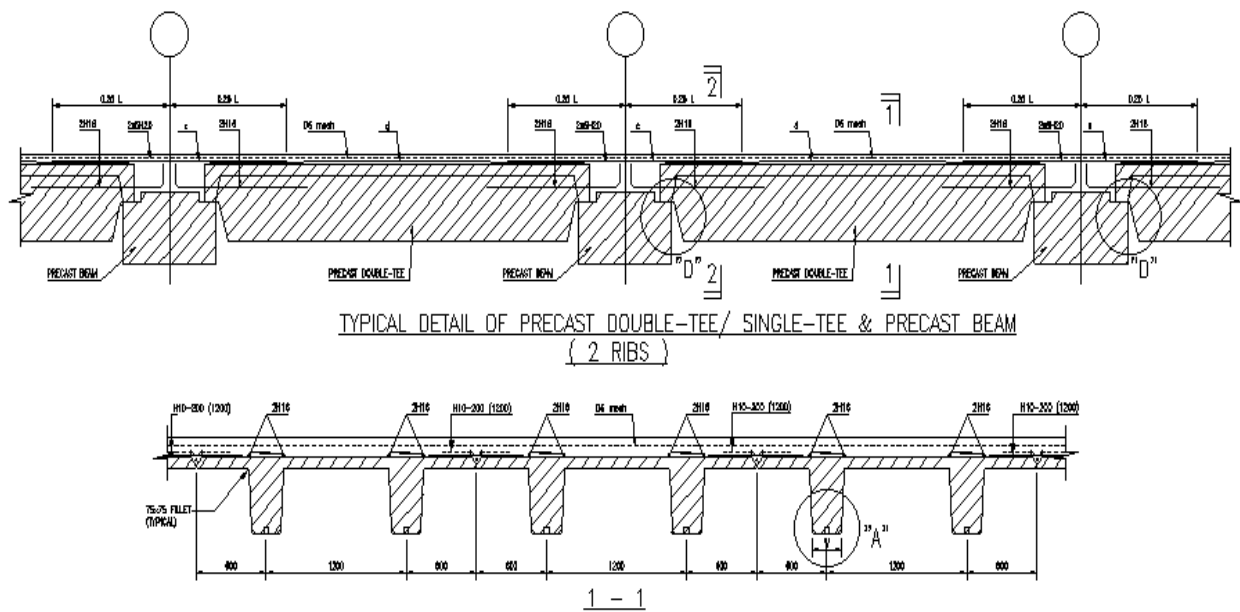


8.2 **Waffle Slab Round Opening**

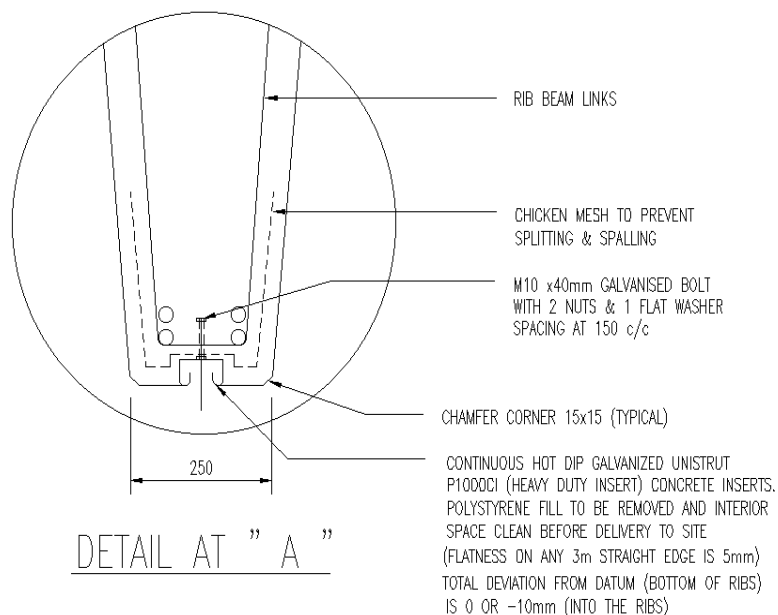


GC to provide temporary cover for all waffle slab precast or cast-in-situ openings. The temporary cover should be a load bearing and can withstand the crane load requirements in cleanroom level during construction period.

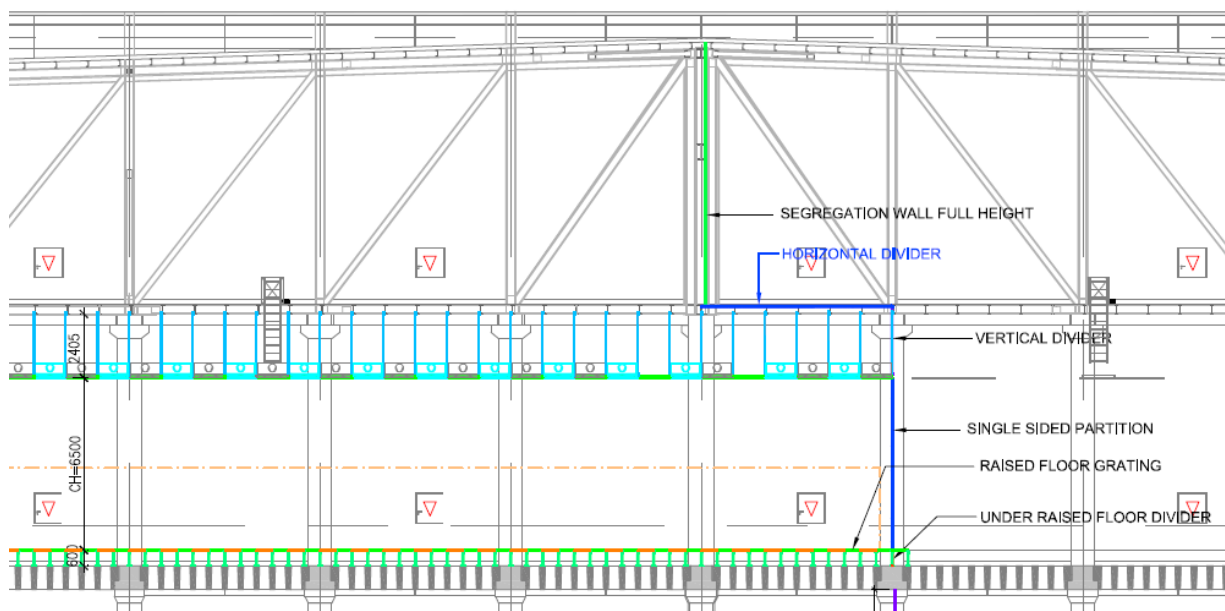
8.3 **Cleanroom Double Tee Slab Typical Details With Uni-strut**

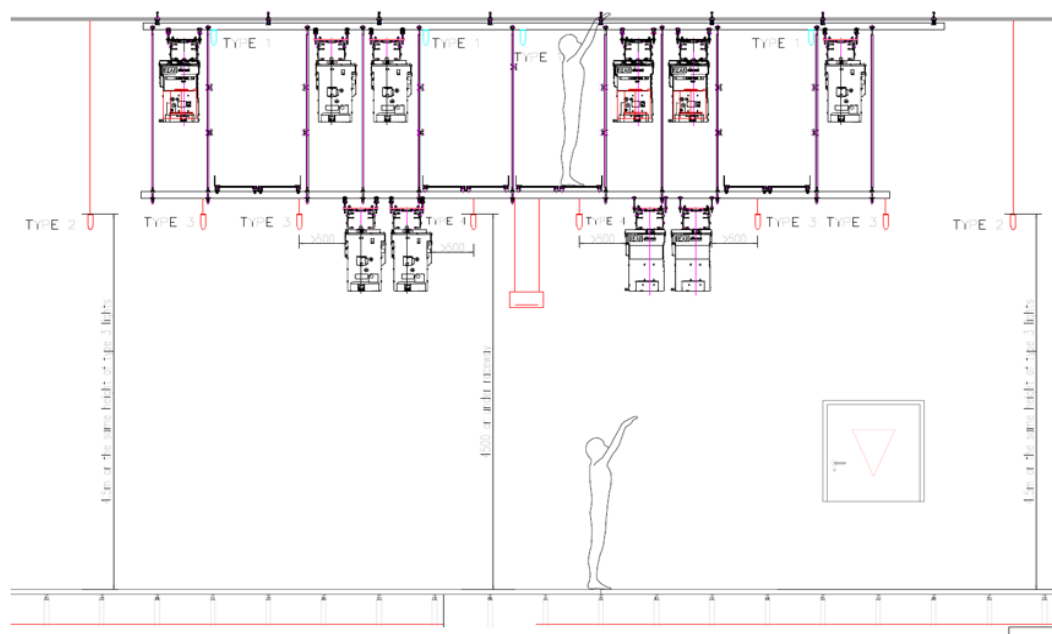


To provide uni-strut at the soffit of each DT/ST rib as shown in the details for all ceiling services instead of drilling works on site. Uni-strut catalogue and maximum load should be provided for Micron Approval.

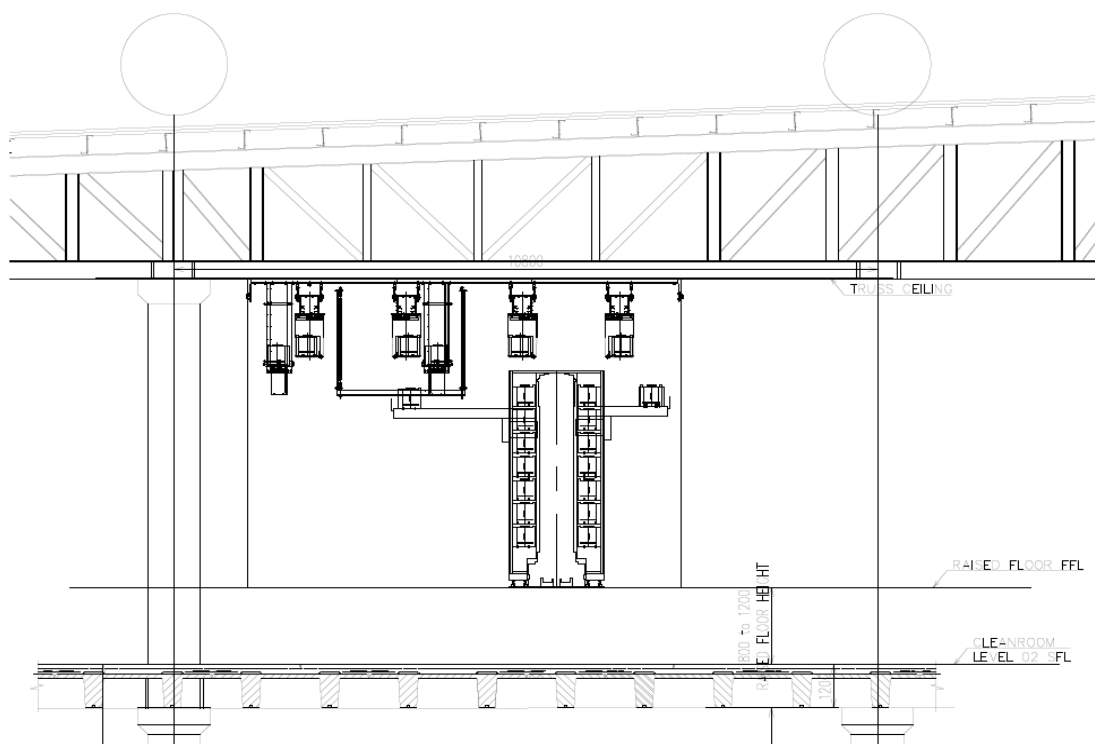


8.4 **Cleanroom Typical Height Requirements Including Non-EUV.**

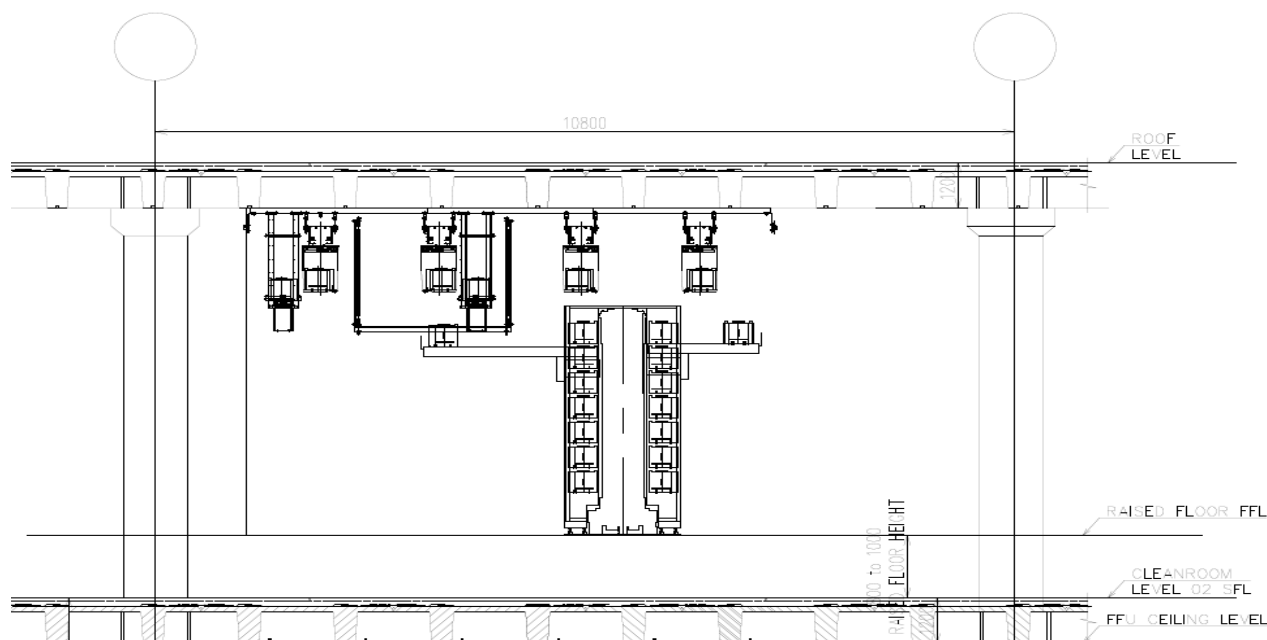




Section A - Typical Front-End FAB Height Requirements with Steel Roof Truss and 2-level AMHS, for reference only



Section B - Typical Assembly & Test (A&T) with Steel Roof Truss, for reference only



Key Considerations for Building design with AMHS

- 1
 - Install Zero Footprint Storage (ZFS- STB/UTB) allow material to be stored and hence allow more tools to utilize cleanroom space.
 - Maximum height of carrier (eg: FOUF/FOSB~ 338mm) + clearance for the vehicle to move along determine how low the UTB need to be.
 - To install both UTB and STB, UTB need to be slightly lower than AMHS design without STB. Minimum 4.8m ceiling height preferred.
- 2
 - >2.7m above floor ensure OHT able to continue to service without disruption when there is any tool maintenance or tool movement. *(this value depends on the tallest tool in production ~eg: Devoid oven for CR2)*
 - Height limitation need to consider maintenance activity such as lifting of cover, stepping on platform will not disrupt OHT operation during manufacturing operation
- 3
 - Considering SEMI S2 overhead clearance height, it is a necessary to have a minimum height of 2.48m
 - Ceiling's height can be lower If tallest equipment is below 2.48m, at >4.58m

Section C - Typical Assembly & Test (A&T) with RC Roof Slab, For Reference Only

8.5 **Metal Roof Truss (Top & Bottom Chord) Loadings and Deflection**

8.5.1. Refer to below tabulated SDL and LL for top chord.

TOP CHORD		
1. Metal Roofing Sheet	0.075 kN/m ²	SDL
2. Insulation (50mm)	0.040 kN/m ²	SDL
3. Sisalation Foil	0.005 kN/m ²	SDL
4. Roof Mesh	0.005 kN/m ²	SDL
5. M&E Services	1.600 kN/m ²	SDL
6. Purlins @ 1.2m Spacing	0.220 kN/m ²	SDL(Varies depends on Purlins Spacing)
7. Roof (Top Chord) Live Load	0.750 kN/m ²	(Refer to the Local Live Load Codal Requirements)

8.5.2. Refer to below tabulated SDL and LL for bottom chord.

BOTTOM CHORD		
1. Ceiling Panel	0.060 kN/m ²	SDL
2. Filter + Housing	0.163 kN/m ²	SDL
3. Ceiling Grids	0.048 kN/m ²	SDL
4. Suspension Rods	0.014 kN/m ²	SDL
5. M&E Services	0.900 kN/m ²	SDL
6. Catwalk/Gratings	0.380 kN/m ²	SDL
7. Water Transport System	1.000 kN/m ²	SDL
8. Ceiling Purlins @ 1.2m Spacing	0.280 kN/m ²	SDL (Varies depends on Purlins Spacing)
9. Roof (Bottom Chord) Live Load	5.000 kN/m ² (Front End) 1.500 kN/m ² (Back End)	(Refer to Section 6.2.1 & 6.2.2 for AMHS Loading Tabulation for Back End and Front End)

* For Front End AMHS Live Load of 500 kg/m² or 5 kPa, 70% of total load is DL=3.5 kPa and 30% is LL=1.5 kPa or moving loads.

*Cleanroom Service Ceiling to include as additional Live Load.

***Cleanroom Vendor Muratec's Ceiling Loading Tabulation**



Loading Condition Calculation Condition

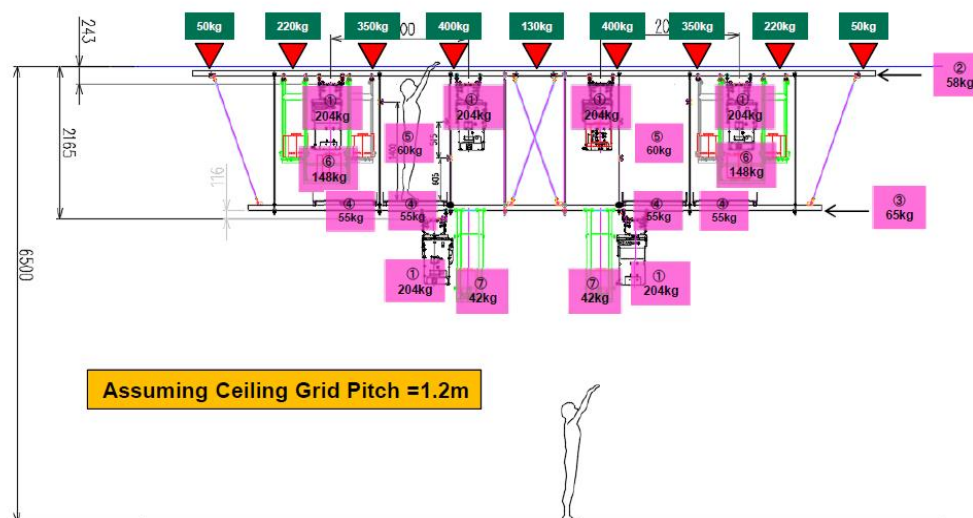
No	項目	詳細	荷重/1.2m	合計 (/1.2m)
①	Vehicle + Rail (G3-3X)	Vehicle+FOUP Rail (Including Hanger + Raceway)	150kg 54kg	204kg
②	1 st Layer Raceway +Bracket to attach on grid	1 st Layer Raceway : 4.3kg/m Bracket to attach on grid : 2kg/pcs	Depend on Layout	
③	3 rd Layer Raceway	3 rd Layer Raceway : 4.3kg/m Hanger rod : 2.3kg/pcs Brace : 2kg/pcs	Depend on Layout	
④	Catwalk	SUS 70% Opening Grating 900W Raceway Hand Rail, Toe Guard	15kg 16kg 14kg	2 Row : 55kg 3 Row : 71kg 4 Row : 100kg
⑤	Person	Person (only during maintenance work)	60kg/person	
⑥	Highway OHB	OHB + FOUP	148kg	148kg
⑦	Interface OHB	OHB + FOUP	42kg	42kg
⑧	Vehicle + Rail (SRC340)	Vehicle + FOUP Rail (Including Hanger + Raceway)	170kg 27kg	197kg
⑨	Interface OHB With IF-Panel	OHB + FOUP + IF-Panel	52	52kg
⑩	Middle Platform	Hanger/Raceway/Catwalk/Ladder	Total load=100kg (supported by 2 raceway)	50kg/raceway

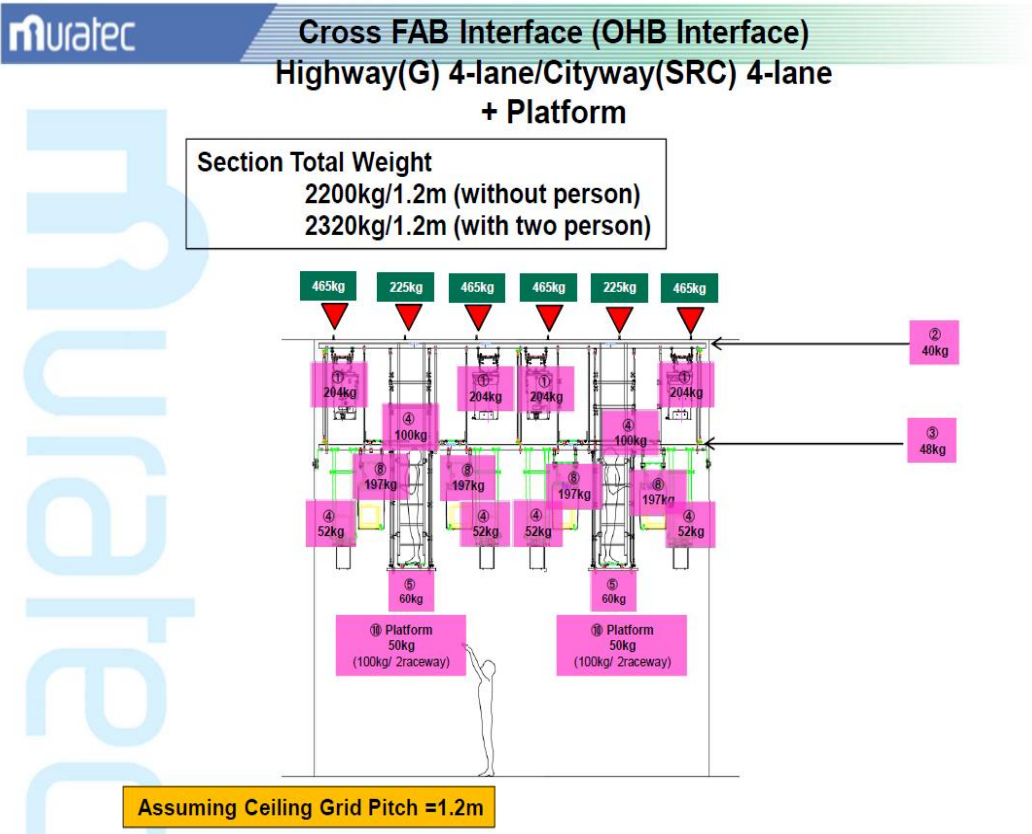
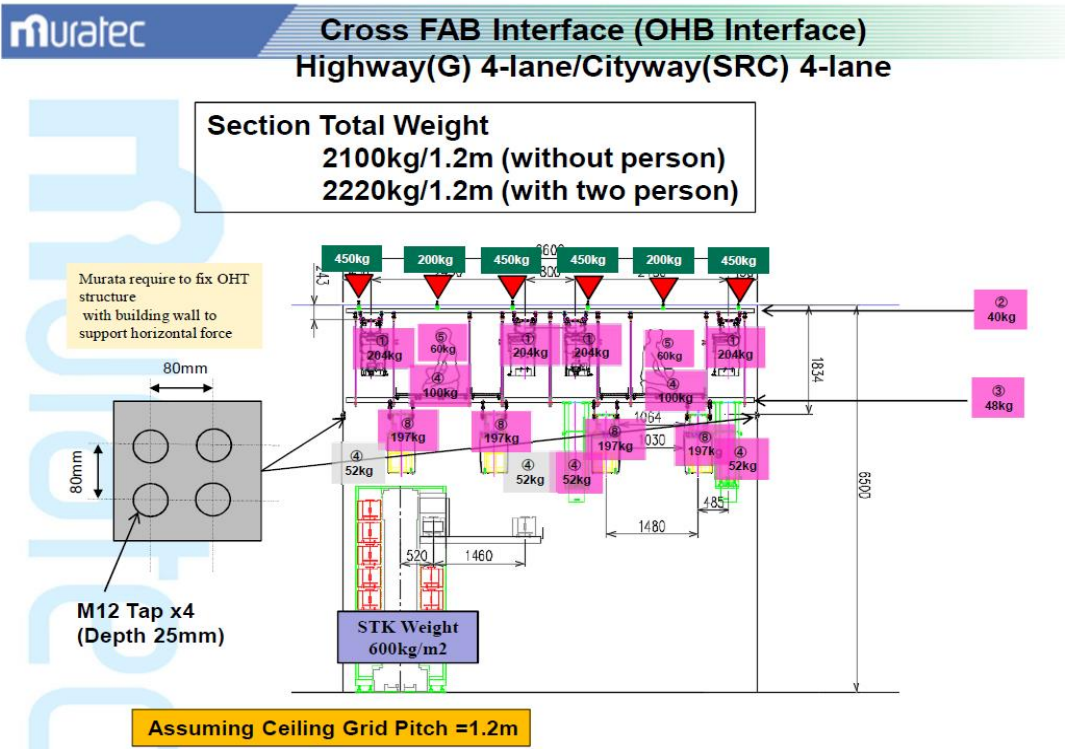


F10A Typical Intra Section

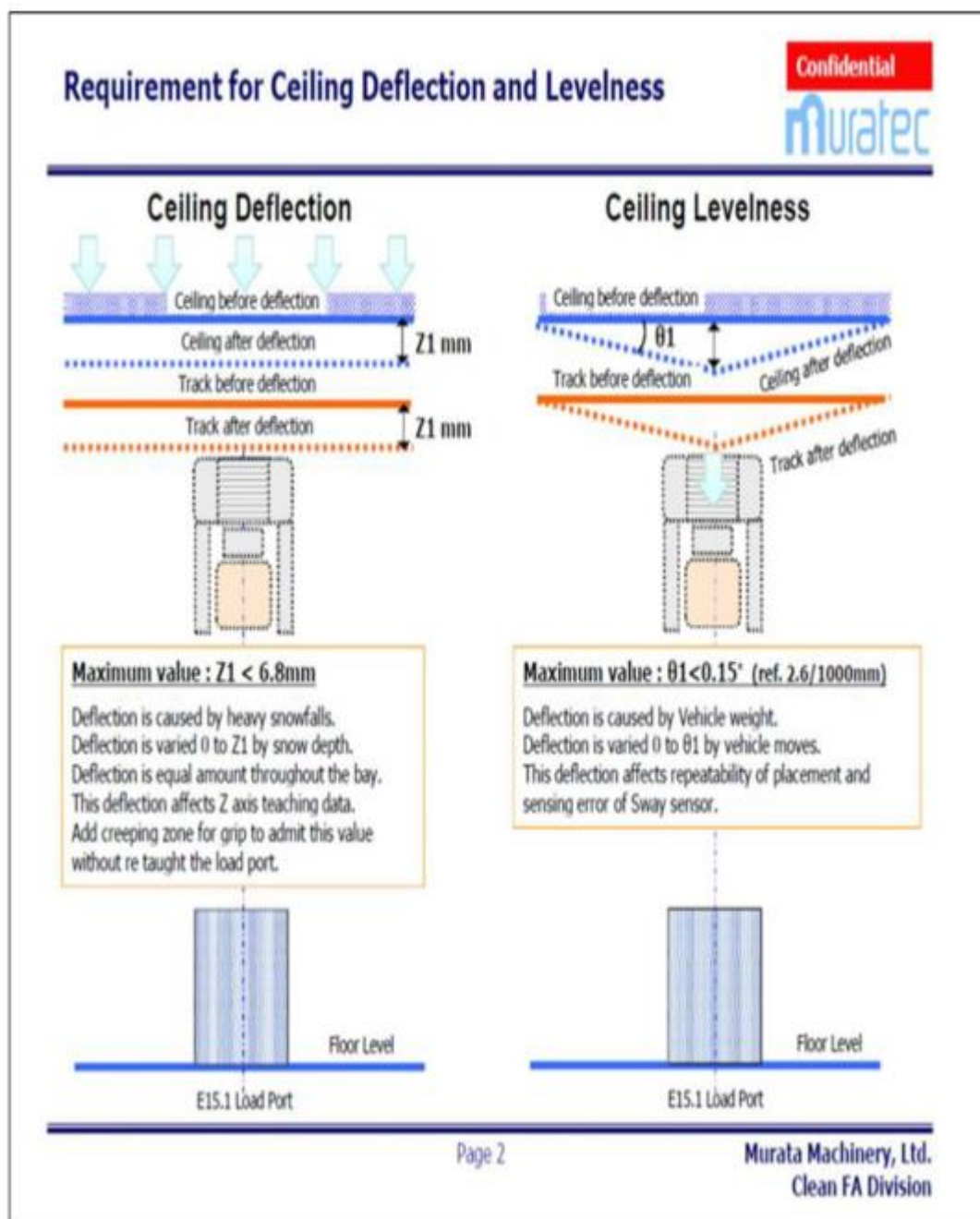
Section Total Weight

1950kg/1.2m (without person)
2070kg/1.2m (with two person)

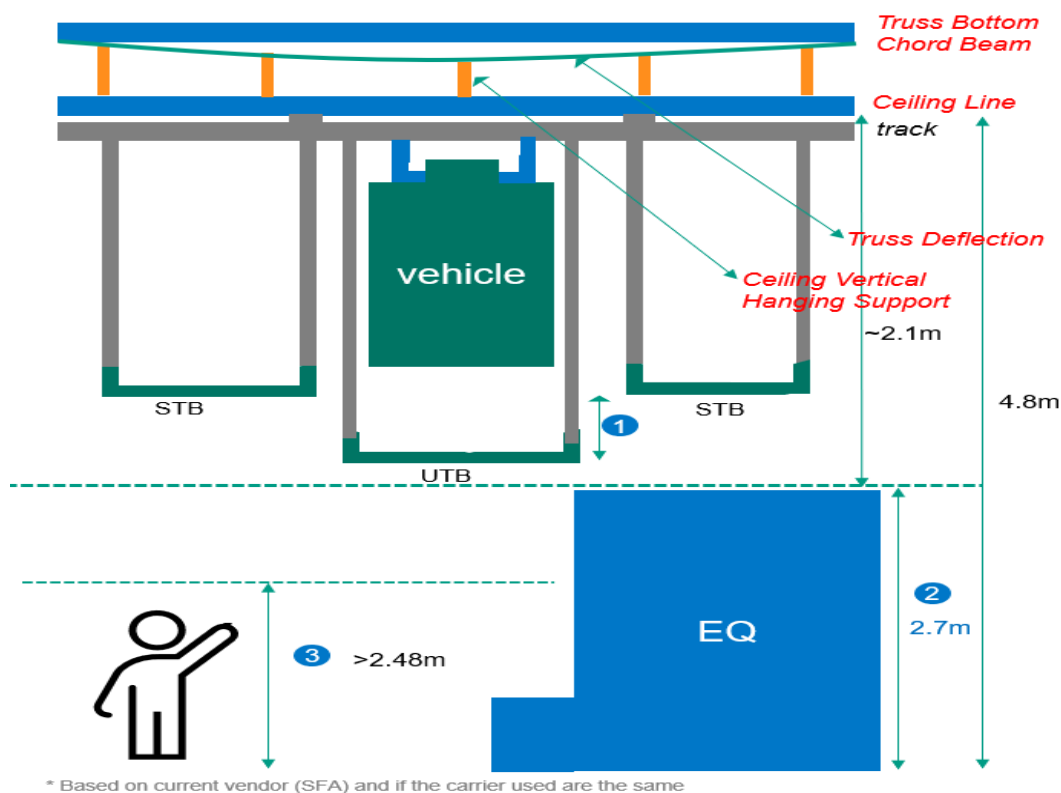




8.5.3. Front End Deflection of Truss Bottom Chord Requirement as per Cleanroom Vendor Muratec In FAB Location with Snow Loads.



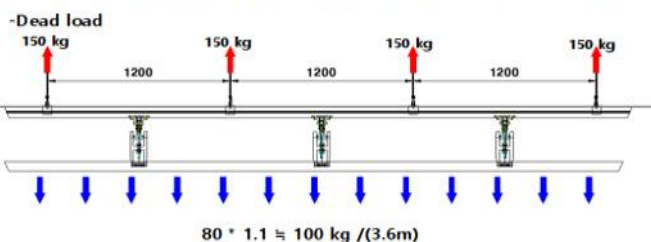
8.5.4. Back End Deflection of Truss Bottom Chord Requirement as per Cleanroom Vendor SFA. The total level/vertical deflection of the ceiling is within 10mm. Ceiling should be within 1mm/m in OHT direction and left/right surface straightness.



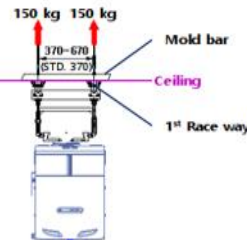
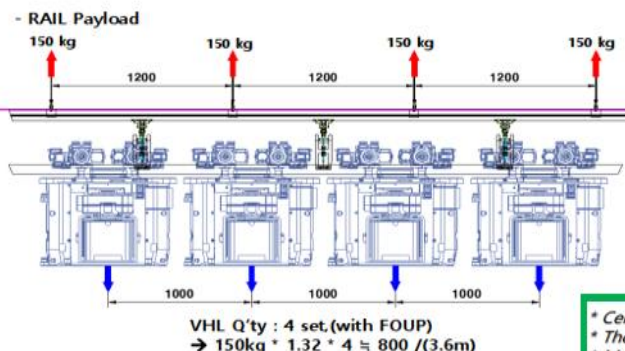
Standard Installation Load (Track + Vehicle)

Configuration

◆ Semiconductor OHT ceiling installation load_1 stage (static load)_3.6m, based on 8EA fastening points



- ▼ Static load per ceiling fixing point
 - Fastening point quantity per 3.6 M : 8 EA
 - Point Load = (Dead Load + Payload)/Point Quantity
 - = (100 + 800) / 8
 - ≈ 120kg
- ▼ ceiling installation load = 120 kg ≤ 150 kg



- * Ceiling should be within 1mm/m in OHT direction and left/right surface straightness
- * The total level/vertical deflection of the ceiling is within 10mm.
- * Mold bar section shape required

9.0 Vibration Mitigation Countermeasures

9.1 Actual Performance / Periodic Vibration Monitoring

- Vibration requirements is included in the overall structural design of the FAB Building during concept, 30%, 60%, 90% until IFC drawing submission.
- Vibration specialist is engaged at the beginning of the project and continuously coordinate with structural engineers or A/E to meet Micron vibration requirements. Vibration specialist is under the scope of A/E.
 - Dynamic design analysis for structural system. Stiffened structural waffle slab, double tees, and shear walls to withstand the vibration requirements.
 - Dynamic design analysis for all mechanical equipment including future equipment's.

9.2 If not Meet, Mitigation Guidelines / OCAP

- In the case of existing Micron FAB Buildings or on-going construction of FAB Building where vibration requirements were added after, Micron to engage a vibration specialist and A/E structural consultant to re-assess the existing building structural members whether it can withstand the new vibration requirements or advise Micron for rectification/strengthening of existing structural members. Micron to assess the cost and schedule impact.

9.3 List of Design / Actual Mitigation Guidelines or Plans

- In the case where new FAB is constructed beside or near the Existing Micron FAB Building, to design the piling as micro piles instead of spun or bored piles to have less impact on vibration during piling works.
- During construction, when the vibration monitoring exceeded, construction piling team should stop the works immediately. Reduce the number of piling rigs and continuously monitor the vibration until it reached the required VC.
- Other design proposal to consider.
 - Introduce a rubber/dampener at the AMHS rod or ceiling rod.
 - Anti-vibration pedestal on the impacted individual tools.
 - Thickening of existing beams/slab to stiffen the existing structural members.
 - Design a secondary steel structure to withstand the new loading and vibration requirements. To check the existing columns, beams, and foundation of the buildings if structurally feasible.
 - Enlargement of existing columns or fiber wrap to strengthen due to additional loadings

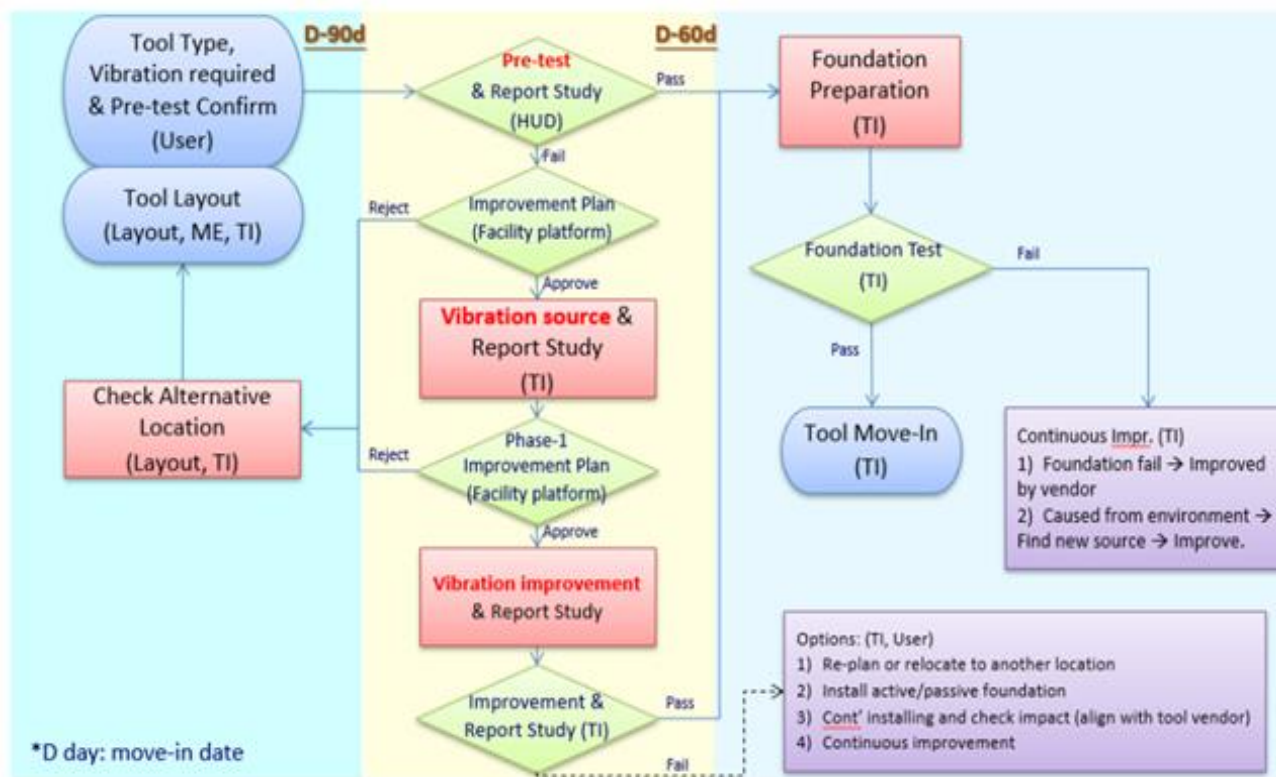
9.4 Vibration Process Flow Check/Monitor During New Equipment/Tool Install

Below the BKM from FW HVM for reference.

ARCI for vibration sensitive equipment

#	Activities	Tool Owner/ME	KEG	Layout (IE)	Tool Install (FAC)
1	Inform tool vibration requirements(KEG Spec Document) for FOAK or re-lay out.	Accountable	Consult	Inform	Inform
2	Provide planned location of tool sit in position	Inform	Inform	Responsible	Inform
3	Conduct and provide vibration test result on tool sit in location	Inform	Inform	Inform	Responsible
4.1	Approve tool sit in location if planned location is <u>accepted</u> by tool owner and KEG	Responsible	Consult	Inform	Consult
4.2	Provide alternate sit in location if planned location is <u>not accepted</u> by tool owner and KEG (Repeat activity #2 & #3)	Inform	Inform	Responsible	Consult
5.1	Foundation preparation (Concrete, Steel,etc)	Consult	Consult	Inform	Responsible
5.2.a	Active damper base plate preparation	Consult	Consult	Inform	Responsible
5.2.b	Active damper preparation	Responsible	Consult	Inform	Inform
6	Foundation/active damper installation	Inform	Inform	Inform	Responsible
7	Tool ODD(Original Deliver Date) and move in	Inform	Consult	Responsible	Inform

Micro-Vibration Test Process Flow

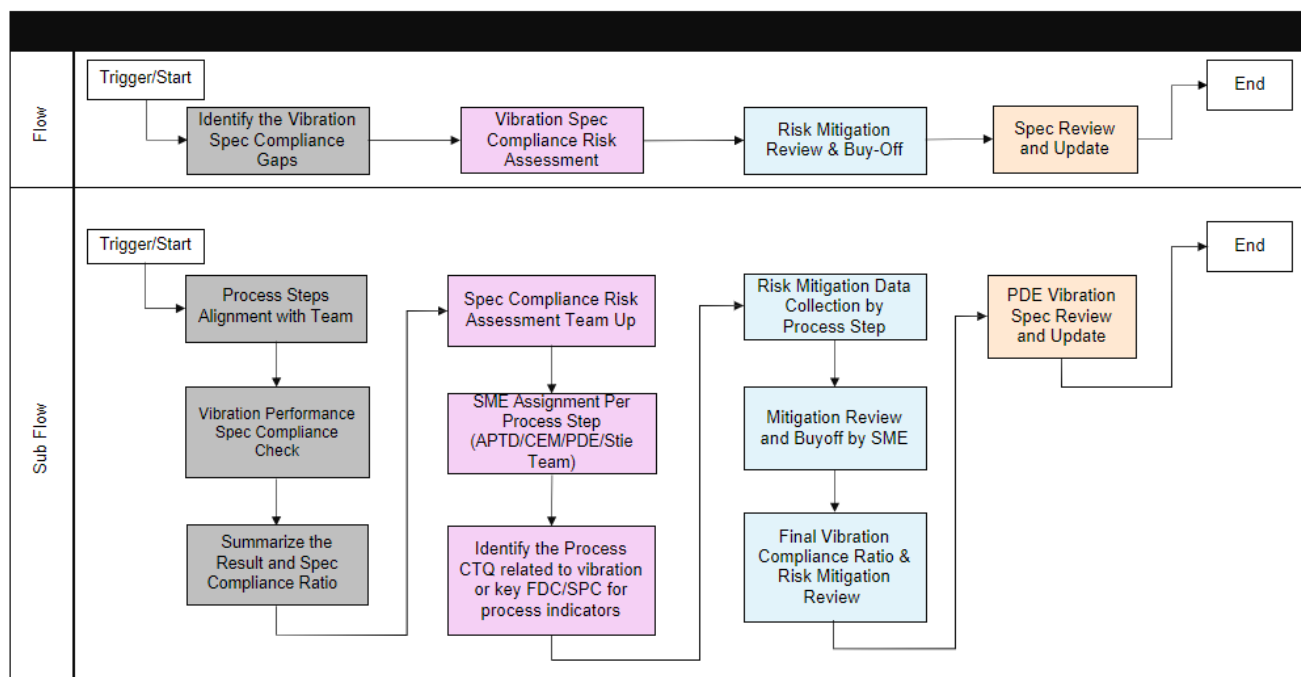


9.5 Vibration Assessment Workflow & ARCI for AT Existing Buildings & Equipment

ARCI

Activity	GQ	PDE KEG	APTD	CEM	Site Team
Identify the Vibration Spec Compliance Gaps					
Process Steps Alignment with Team	I	A/R	R	C	R
Vibration Performance Spec Compliance Check	I	R	C	C	A/R
Summarize the Result and Spec Compliance Ratio	I	R	C	C	A/R
Vibration Spec Compliance Risk Assessment					
Spec Compliance Risk Assessment Team Up	C/I	A : FOAK R : SOAK	R	R	A : SOAK R : FOAK
SME Assignment Per Process Step (APTD/CEM/PDE/Stie Team)	C/I	A : FOAK R : SOAK	R	R	A : SOAK R : FOAK
Identify the Process CTQ related to vibration or key FDC/SPC for process indicators	C/I	A : FOAK R : SOAK	R	R	A : SOAK R : FOAK
Risk Mitigation Review & Buy-Off					
Risk Mitigation Data Collection by Process Step	C/I	A : FOAK R : SOAK	R	R	A : SOAK R : FOAK
Mitigation Review and Buyoff by SME	C/I	A : FOAK R : SOAK	R	R	A : SOAK R : FOAK
Final Vibration Compliance Ratio & Risk Mitigation Review	C/I	A : FOAK R : SOAK	R	R	A : SOAK R : FOAK
Spec Review and Update					
PDE Vibration Spec Review and Update	I	A/R	I	I	I

Vibration Assessment Workflow



10.0 Vibration Design/Monitoring Procedure Before/During/After Construction of New FAB/Production Building

10.1 Bare Ground Vibration Measurement & Preliminary Design

- Perform bare ground vibration measurements/monitoring for comparison with Micron required specifications. Measurements to be performed according to IEST standards.
- Submission of complete and final report.
- Review the concept design to achieve project criteria, performing first calculations for waffle slab/CIS (vertical and horizontal stiffness) to check the compatibility of requirements.

10.2 Continuous Vibration Monitoring at the Existing Cleanroom During Test Piling

- Mobilize and install a PC based continuous monitoring system at the existing production building at waffle slab level (one system with one triaxial sensor block)
- Continuous alert
- Retrieve data during the monitoring period for processing
- Submission of alarm summary report
- Demobilize system at end of monitoring period

10.3 Dynamic Design Analysis for Structural System

- Perform detailed FEM dynamic analysis for the proposed structural system, to determine the natural frequencies and dynamic stiffness for suitability to achieve specified VC vibration levels.
- Evaluate structural requirements for foundation, Level 1, Level 2 slab, Level 3, and Upper Levels, Waffle slab, shear wall thickness, column sizes and spacing including the column supporting the roof and support wings, pipe racks and etc
- Based on analysis, an assessment to be made of the proposed design for the VC criteria specified.
- Submission of formal report.

10.4 Dynamic Design Analysis for Mechanical Equipment

- Perform the necessary dynamic analysis for mechanical equipment, piping and racks, Internal piperacks at Level 1, Level 2, and other subfab levels to facilitate compliance with specified VC requirements and propose necessary mechanical vibration isolation.
- Submission of formal report.

10.5 Continuous Vibration Monitoring at the Existing Building Cleanroom During Construction Stage (During New Building Piling, Sub and Superstructure Works)

- Mobilize and install a PC based continuous monitoring system at the existing production building at waffle slab level (one system with one triaxial sensor block).
- Continuous alert.
- Retrieve data every fortnightly during the monitoring period for processing.
- Submission of fortnightly alarm summary report.
- Demobilize system at end of monitoring period.

10.6 Commissioning Measurement

- Perform commissioning structural vibration measurement on a random, representative sample of locations on Cleanroom Level waffle slab of the new Fab with all utilities running at normal conditions, when the Fab is ready for equipment (RFE).
- Compare measured results with specified VC limits.
- Submission of Formal/Final reports.

10.7 Sealant

- Sealant and Coating finishes for Cleanroom floor level should be compatible with cleanroom requirements. Contractor to submit the test report for Micron review and approval.

11.0 Cleanroom Interior Fit Out

11.1 Cleanroom Ceiling System

11.1.1 Description

11.1.1.1 The Cleanroom Ceiling System shall consist of:

1. Ceiling grid framework
2. FFUs with ULPA filters
3. Walkable blank ceiling panels
4. Cleanroom lighting fixtures
5. Sprinklers

11.1.1.2 The cleanroom ceiling system shall be designed according to ISO 14644. The ceiling grid shall be based on a 1,200mm x 1,200mm standard size system.

11.1.1.3 The FFUs and blank panels shall be installed onto the cleanroom ceiling grid framework, which is suspended from the building structure.

11.1.1.4 All ceiling grid structure materials shall be made of aluminum. Utility penetrations such as for cabling for lighting fixtures, sprinkler pipes, etc. can be provided at the intersectional pieces or at the blanks if required. The location of the sprinkler heads will be designed in full compliance to the local authority requirements. The openings or cut-outs of ceiling panels for equipment, ducts or pipes shall be avoided to maintain the integrity of ceiling system.

11.1.1.5 The filter-ceiling framework shall offer full flexibility with respect to the location of cleanroom partitions, FFUs, blank panels, and lighting fixtures.

11.1.1.6 Coordinate the cleanroom partition locations with sprinkler protection arrangement to suit local regulations.

11.1.1.7 The lighting fixtures shall achieve the required lighting level and are mounted in the ceiling grid in all areas. The lighting fixtures can be shifted to every individual ceiling grid. The light fixtures shall be coordinated with the AMHS system and all other utilities to ensure adequate lighting levels in the Cleanroom.

11.1.1.8 The finished grid system shall be accessible for personnel traffic. In case that FFUs or blank panels must be exchanged or replaced, this can be done from the plenum above the cleanroom or from below the cleanroom ceiling.

11.1.1.9 All blank panels shall be "walkable" throughout entire ceiling grid. Modified blank panels which are no longer certified to be "walkable" must be identified, clearly

indicated and/or barricaded to prevent personnel walk onto these panels.

11.1.1.10 The entire ceiling profile system facing the cleanroom shall be anodized aluminum.

11.1.1.11 To access the cleanroom ceiling, FFUs, make-up air units, and facilities installations, a catwalk system or a grated platform shall be installed. The material shall be galvanized steel. Where necessary for safety reasons, handrails shall be installed at the boundary of the walkway; ladders shall be provided for access to the cleanroom ceiling level.

11.1.2 Cleanroom Ceiling Quality Characteristics

11.1.2.1 Ceiling Grid Framework

1. Ceiling Grid size shall be 1,200mm x 1,200mm or 600mm x 1,200mm
2. Framework material shall be anodized extruded, aluminum 'T'- channel profiles
3. Sealant shall be elastic rubber gasket
4. Hanging rods shall be galvanized steel

11.1.2.2 Cleanroom lighting shall be either pendant type hanging from ceiling grid, or tear-drop type, mounted on ceiling grid to prevent disturbing the air flow from the FFUs on top of the ceiling system. The lighting location shall be coordinated with AMHS, ductwork, piping, and other utilities to ensure adequate lighting levels in the Cleanroom areas.

11.2 Cleanroom Walls

11.2.1 Description

11.2.1.1 The cleanroom partition wall system consists of Modular single sided and/or double-sided walls, static dissipative type.

11.2.1.2 The cleanroom partition wall system shall be installed at the perimeter of the Fab, covering the entire Cleanroom area and the Cleanroom Support Rooms.

11.2.1.3 The wall system shall be designed to match the given grid dimension of the raised floor and the grid dimensions of the filter ceiling, wherever possible.

11.2.1.4 The wall panels shall be easily removable and interchangeable through its modular construction design.

- These panels shall be removable and re-usable without damaging their construction or material.
- 11.2.1.5 The partition walls shall be static dissipative and have no acoustic infill.
- 11.2.1.6 The cleanroom partitions shall be stable and rigid and shall not be affected by penetrations. They shall also have sufficient stiffness against bending. When de-mounting panels or elements, the stability of the remaining partitions will be neither reduced nor affected.
- 11.2.1.7 The partition system shall consist of a vertical and horizontal framing (stud and crossbar) of tubular or channel-type profiles onto which coated sheet metal panel (single-sided wall) or honeycomb panel (double-sided wall) to be installed. Door and window frames shall be incorporated in the system. All joints between elements and framework shall be completely sealed.

11.2.2 Cleanroom Wall Quality Characteristics

11.2.2.1 Single / Double-Wall Panels

Board Material:	Single side Cleanroom wall, metal sheet, thickness of sheet metal 1.0 to 1.5 mm Double side Cleanroom wall is honeycomb filling covered with 1.0 mm steel sheets on both sides
Finishing:	Conductive, coil-coated or wet paint, min. 0.025mm dry thickness.
Static Dissipative Appearance:	1 to 100 MΩ
Panel module:	Typically, 600 or 1200 mm
Surface:	Smooth, cleanable, impact resistant, watertight, maintenance-free, lightweight, non-particle generating nor shedding when rubbed against or due to air movement

11.3 Cleanroom Flooring System

11.3.1 Description

11.3.1.1 The raised floor system shall consist of:

1. Solid floor tiles including a static dissipative vinyl covering
2. Perforated or Grated floor tiles including a static dissipative vinyl covering
3. Pedestals
4. Insulation plates
5. Stringers
6. Bracing or stiffeners (where required)
7. Raised floor separations
8. Where applicable, Type 2 Raised Floor System with underfloor substructure frame consisting of primary beams, secondary beams, stringers, adapter pedestals, and mini pedestals

11.3.1.2 A raised floor system, combined from solid panels, grating panels, and perforated panels (free cross sectional opening of perforation approximately 17-35%, depends on the design criteria) shall be installed covering the entire cleanroom production area.

11.3.1.3 The complete raised floor shall be installed with an aluminum stringer without any opening or cut-outs for equipment, openings and cutouts shall be provided during the "Tool Install" phase. Heavy equipment and equipment with tight vibration specifications shall be supported by special tool foundations / tool pedestal bases, which shall be installed directly upon the concrete waffle slab (typically during the "Tool Install" phase).

11.3.1.4 The grid of the raised floor shall be based on a 600mm x 600mm standard size system and shall be aligned with the centerlines of the Cleanroom ceiling grid. The floor tiles shall be completed with a layer of conductive vinyl covering.

11.3.1.5 Perforated and Grating floor tiles shall be provided to allow re-circulating air to pass through to the clean subfab area, which is utilized as the Cleanroom air return path.

11.3.1.6 Airtight separation shall be installed from the concrete slab to the underside of the raised floor panel to separate rooms having different temperature and pressure criteria.

11.3.1.7 The entire raised floor pedestals shall be powder coated off-site, for example, factory coated. All raised floors shall be laterally stable in all directions to ensure a solid under-structure so that the raised floor panels can be taken out or shifted easily. The floor tiles shall be supported at each corner by height - adjustable pedestal heads. These pedestals shall be secured to the finished concrete slab.

11.3.1.8 Insulation plates shall be placed on each pedestal head to attenuate footstep sound and assure electrical conductivity. The raised floor shall be connected directly to the perimeter concrete slab structure without any isolation, i.e., expansion joint. At the perimeter, the raised floor shall be installed with bracing to ensure stability after the initial installation.

11.3.2 Raised Floor Quality Characteristics

11.3.2.1 Raised Floor

Type:	Standard Raised Floor System
Load Capacity:	min. 245 N/cm ² (350 psi), or TBD based on project requirements
Concentrated Load:	1500 kgf (3300 lbf), or TBD based on project requirements
Max. Deflection:	1/300 (2 mm) maximum of length tile concentrated load, or TBD based on project requirements
Height:	TBD based on project requirements

Type:	Heavy Duty Raised Floor System
Concentrated Load:	3000 kgf (6600 lbf), or TBD based on project requirements
Max. Deflection:	1/300 (2 mm) maximum of length tile concentrated load, or TBD based on project requirements
Height:	TBD based on project requirements

11.3.2.2 Floor Panels

Material:	True milled die-cast aluminum, not coated, with very high accuracy. Perforations shall be factory drilled and subsequently thoroughly cleaned before packing. Perforation to be calculated as per the airflow rate of approximately 500 to 860 m3/h at pressure drop of 5 to 10 Pa
Conductivity:	Electrical Resistance: Average of 1 MΩ between an electrode placed on floor and a ground connection when tested according to ASTM F150 Static Generation: Less than 300 volts when tested according to AATCC-134 at RH 20% Static Decay: 5,000 to 0 volts in less than 0.25 seconds when tested according to FED-STD-101/4046, at RH 15%
Dimensions	600 x 600 mm with very narrow tolerances

11.3.2.3 Pedestals

Material:	Galvanized Steel, installed upright true vertical with steel base plates with holes for fixing the pedestal to concrete slab in special areas, or powder coated, coating thickness of 2mil (0.060 mm) min.
Fine Adjustment:	Threaded rod member and an elevating nut
Height:	TBD based on project requirements
Stringers:	Stainless steel or aluminum, bolt-on type

11.3.2.4 Bracing/Stiffeners

Material:	All aluminum or steel (cadmium), galvanized or powder coated, coating thickness approximately 2mil (0.060 mm) min.
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11.3.2.5 Type 1 System – Underfloor Pedestals and Stringers

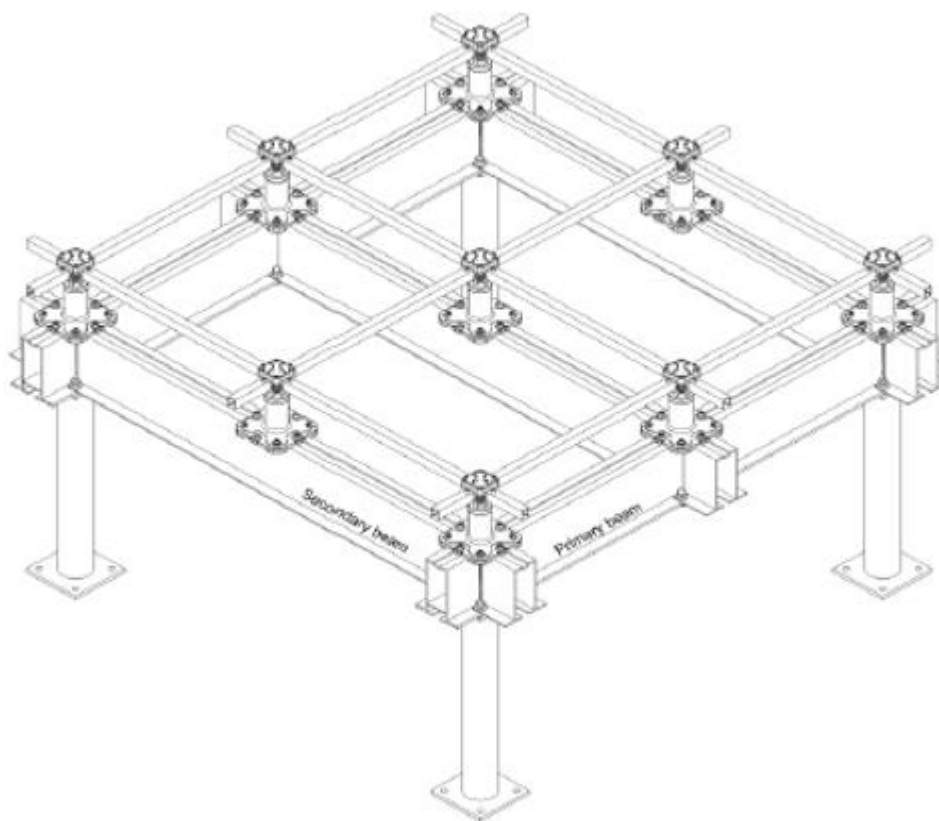
Reference Design for 3,000lb System Raised Floor System



11.3.2.6 Type 2 System - Underfloor Substructure Frame System with Primary Beam, Secondary Beam, Stringer, Adapter Pedestals, and Mini Pedestals

Reference Design for 3,000lb Raised Floor System

Member	Concentrated Load	Ultimate Load	Overturning Moment	Safety Factor	Reference(*)
Mini Pedestal	5,000lbs (2,268kgf)	12,500lbs (5,670kgf)	-	2.5	4. Design Loads, a. Pedestals, (1) Centered Axial Load : 5,000lbs. Ultimate strength of 2.5 times design load.
Secondary Beam	3,000lbs (1,361kgf)/ 0.08inch(2.03mm)	4,500lbs (2,041kgf)	-	1.5	5. Underfloor Adaptor Structure, a. Support Beams, (1) Secondary Beam : 3,000lbs concentrated load with maximum 0.08inch or L/600 deflection, whichever is smaller. (3) Factor of Safety : 1.5 for beams.
Primary Beam	4,000lbs (1,814kgf)/ 0.08inch(2.03mm)	6,000lbs (2,721kgf)	-	1.5	5. Underfloor Adaptor Structure, a. Support Beams, (2) Primary Beam : 4,000lbs concentrated load with maximum 0.08inch or L/600 deflection, whichever is smaller. (3) Factor of Safety : 1.5 for beams.
Post	20,000lbs (9,072kgf)	-	5,000in-lbs (5,761kgf-cm)	-	5. Underfloor Adaptor Structure, b. Support Beam Pedestal Axial Load : 20,000lbs. c. Moment capacity in any direction: 5,000in-lbs.



11.3.3 Resilient Vinyl Flooring System

11.3.3.1 The conductive vinyl flooring system shall consist of:

1. Vinyl sheet (PVC)
2. Vinyl tiles in size 600 x 600 mm or sheet flooring
3. Skirting on the perimeter (if building wall is connected)
4. Static dissipative vinyl flooring shall be applied in sheet form in areas directly in functional connection to the cleanroom, for example, clean staircase, etc.
5. Proper floor leveling shall be achieved by means of a self-leveling compound.

11.3.3.2 Flooring Material

Resilient Vinyl Flooring:	Static dissipative vinyl flooring, concentrated load carrying capability 800N/cm ² the entire flooring system without damaging the substrate
Conductivity:	Floor to ground resistance between 25KΩ and 1MΩ, Test voltage to be 500 volts DC. This is achieved by conductive graphite ingredients and conductive glue in connection with copper earthing tape linked to the overall earthing system
Size:	600 x 600 mm Tile
Overall Thickness	2.0 mm
Wear Layer	2.0 mm
Weight	Approximately 2.8 kg/m ²
Electrical Resistance	Electrical Resistance: Average of 1 to 1,000 MΩ between an electrode placed on floor and a ground connection when tested according to ASTM F150
Seams	Welding required
Color	To be selected from manufacturer's standard color line

12.0 Document Control

11.1 Approval: GLOBAL_FAC_GFTT_DOC_APPR

11.2 Notification:

GLOBAL_FAC_MANAGERS
GLOBAL_FAC_COMMUNICATIONS
GLOBAL_FAC_CONSTRUCTION
GLOBAL_FAC_NOTIFY
FCG_F4_FAC_NOTIFY
FCG_F6_FAC_NOTIFY
FCG_F10_FAC_NOTIFY
FCG_F11_FAC_NOTIFY
FCG_F15_FAC_NOTIFY
FCG_F16_FAC_NOTIFY
FCG_MMP_FAC_NOTIFY
FCG_MMY_FAC_NOTIFY
FCG_MSB_FAC_NOTIFY
FCG_MXA_FAC_NOTIFY
FCG_MTB_FAC_NOTIFY

11.3 Notification:

All Micron Team Members at all Sites

11.4 Retention

Adherence to the requirements specified in the
[Global Facilities – Document Control Procedures](#) and
[Global Records Retention Policy](#)

11.5 Review

Biennially (two years)

12.0 Revision History

Rev. No.	Section Changed	Details of Changes	Date
0	NA	New Document	09/08/2022
1	5.0 6.0 7.0 8.0 9.0	Additional vibration control information. 1. Additional A&T AMHS Loading and Vehicle Speed 2. Additional Front End AMHS Loading 3. Additional EUV Equipment Layout and Section 4. Additional EUV HiNA Crane Loads 1. Updated Back End Vibration Requirements 2. Additional EUV Tools Equipment Dynamic Stiffness Vibration Requirements 1. Updated Waffle Slab Future Opening Requirements - 3 Module 2. Updated AMHS Cleanroom Section for Front and Back End 3. Updated the Truss Top and Bottom Chord Loadings 4. Additional AMHS Bottom Chord Truss Deflection Criteria Additional Vibration Mitigation Countermeasures, RABID for Vibration Sensitive Equipment, and Micro-vibration test process flow.	11/18/2022
2	6.0 and 7.0 9.4	Updated the VC-D+ vibration requirements and location in English conversion. Replacement of ARCI for vibration sensitive equipment.	03/13/2023
3	8.5.2 8.5.4 9.5	Cleanroom Vendor Muratec's Ceiling Grid Loading Tabulation Back End Deflection Criteria Add in chapter title "9.5 Vibration Assessment Workflow & ARCI for AT Existing Buildings & Equipment" and add in two charts "ARCI" & "Vibration Assessment Workflow".	12/12/2023